

Permutations

Date _____ Period _____

List all possible permutations.

1) ☺, ☼, ♥

☺☼♥ ☼☺♥ ♥☺☼
☺♥☼ ☼♥☺ ♥☼☺

2) ☺, ☼

☺☼ ☼☺

3) 1, 2

12 21

4) A, B, C, taken two at a time

AB BA CA
AC BC CB

Evaluate each expression.

5) ${}_8P_4$

1,680

6) ${}_8P_5$

6,720

7) ${}_7P_5$

2,520

8) ${}_{10}P_3$

720

9) ${}_{10}P_5 - 5$

30,235

10) ${}_{10}P_3$

720

Find the number of unique permutations of the letters in each word.

11) EVERY

60

12) SOLO

12

13) BANDING

2,520

14) STRUTS

180

15) THERMOMETER

831,600

16) BILLIONAIRE

3,326,400

Critical thinking questions:

17) Name a season of the year, where if you rearrange the letters there are 360 unique permutations.

SUMMER or AUTUMN

18) Simplify ${}_xP_{x-2}$

$\frac{x!}{2}$